LAB VENUE:
DIGITAL ELECTRONICS LAB @ E4-03-07

LAB SCHEDULE:

Attend your official lab session. Swapping is not allowed.		
Week	**Activity**	**Graded Component (60%)**
Week 1 10 Aug 2026	No Scheduled Lab	-
Week 2 17 Aug 2026	No Scheduled Lab	-
Week 3 24 Aug 2026	**Lab 1:** Introduction to Vivado	Practice Task (2%) *Recommended to complete task in lab 1*
Week 4 31 Aug 2026	**Lab 2:** Combinational Circuits	Practice Task (4%) *Recommended to complete task in lab 2*
Week 5 7 Sep 2026	**Lab 3:** Progress Checkpoint 1/3	Progression Test (Tentatively 5%) *Further instructions will be announced on Canvas*
Week 6 14 Sep 2026	**Lab 4:** Sequential Circuits	Practice Task (Tentatively 7%) *To complete task within 1 week*
Recess Week	No Scheduled Lab	-
Week 7 28 Sep 2026	**Lab 5:** Project Initiation	Practice Task and Project Proposal (3%) *To complete within 1 week*
Week 8 5 Oct 2026	No Scheduled Lab	-
Week 9 12 Oct 2026	**Lab 6:** Project Work	-
Week 10 19 Oct 2026	**Lab 7:** Progress Checkpoint 2/3	Progression Test (12%) *Further instructions will be announced on Canvas*
Week 11 26 Oct 2026	No Scheduled Lab	-
Week 12 2 Nov 2026	**Lab 8:** Progress Checkpoint 3/3	Project Assessment (27%) *Further details will be given in the Project Manual*
Week 13 9 Nov 2026	No Scheduled Lab	-

Progression Tests have the following characteristics:

- Requires the knowledge and debugging skills from the previous Practice Tasks.
- No usage of A.I.
- No usage of VS Code.
- Limited time: No trial-and-error type of coding. Think before you carefully code!

EE2026: DIGITAL DESIGN

Academic Year 2026-2027, Semester 1

LAB 1: Quick Start Guide to Vivado 2025.2, Basys 3 Development Board, and Verilog HDL

HARDWARE COMPATIBILITY:

- Vivado 2025.2 allows for **Basys3 hardware connection** only on Intel or AMD x86/x64 based processors, and using Windows. It will not allow Basys3 hardware connection on ARM based processors (Such as Apple's M1, M2, M3 etc. processors).

- Virtualisation of windows, or non-English character (Eg. Chinese version of windows) Windows, generally have issues with the Basys3 hardware connection.

FOR ALL EE2026 LAB AND PROJECT SESSIONS:

- You are **strongly encouraged to bring to lab, your own Windows laptop with Vivado 2025.2 already installed**. You may still use the desktop PC in lab if you do not have a **windows laptop (With x86/x64 processors)** that can be brought to lab.

- Use the **D:\MyWork** folder for your work if you are using the lab PC. You are required to **delete** all folders within the **D:\MyWork** folder before starting your lab session, and when you leave the lab.

OVERVIEW:

Using a simple Boolean design problem, an introductory approach to the Vivado software used in EE2026 will be covered.

In this lab:

- An introduction to very basic Verilog HDL (Hardware Description Language) is provided.
- The overall process flow of designing, synthesising, simulating and implementing a program is covered.
- Programming Digilent's Basys 3 development board, which features an FPGA from Xilinx's Artix-7 family, is illustrated.

GRADED PRACTICE TASK [CANVAS SUBMISSION REQUIRED]:

It is required to have the Vivado software already installed on your English-Based Windows laptop with Intel / AMD processors **before** the lab session. The Graded Practice Task for lab 1 (GPT-1) is required to be started during your first lab session and preferably completed during the lab session.

Details for GPT-1 are available at the end of this lab manual.

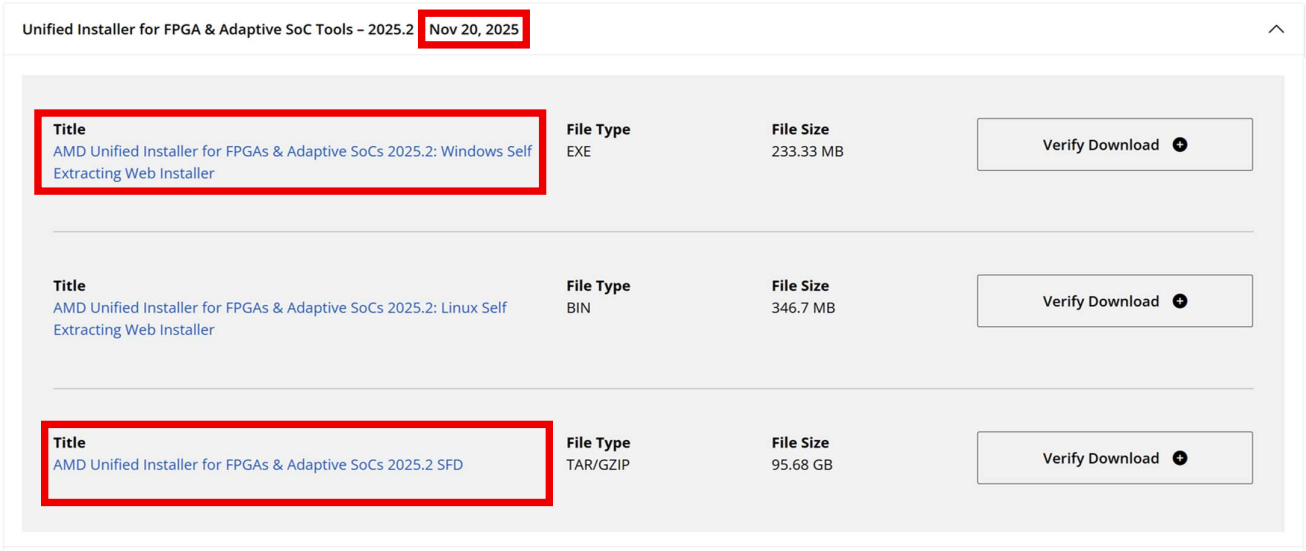
VIVADO SETUP:

The Vivado 2025.2 software is already installed on the computers in the Digital Electronics Lab, and are ready for immediate usage. Some quick guidelines on installing the required software for EE2026 on your personal computer is provided in this section.

Software Download:

AMD Unified Installer for FPGAs & Adaptive SoCs 2025.2 [**Nov 20, 2025**]

Link: **https://www.amd.com/en/support/downloads/adaptive-socs-and-fpgas/development-tools/2025-2.html**

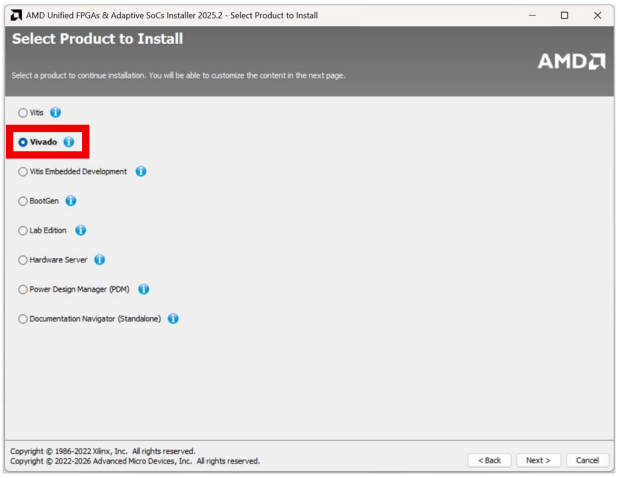


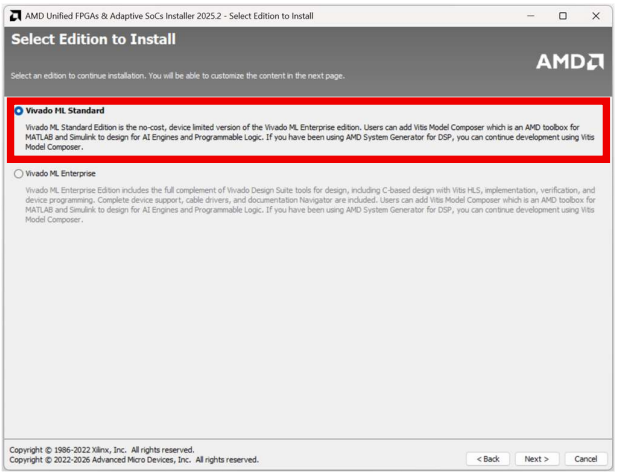
You may choose either one of the two installers. Take note of the file size and the large amount of time you may require (One takes a long time to install while the other one takes a long time to download)!

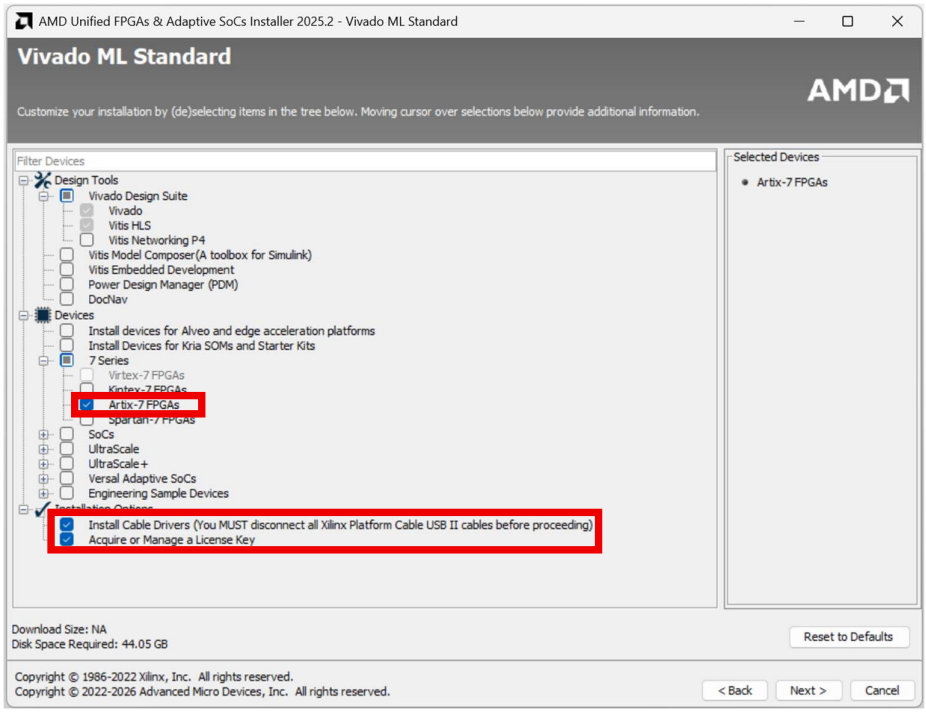
Warning: *Do not use other versions of the software. Only the Vivado 2025.2 Windows OS version should be used. Compatibility issues will occur with other versions of the software, and assessment of your project may not be possible.*

Software Installation

During the installation, select the following options:

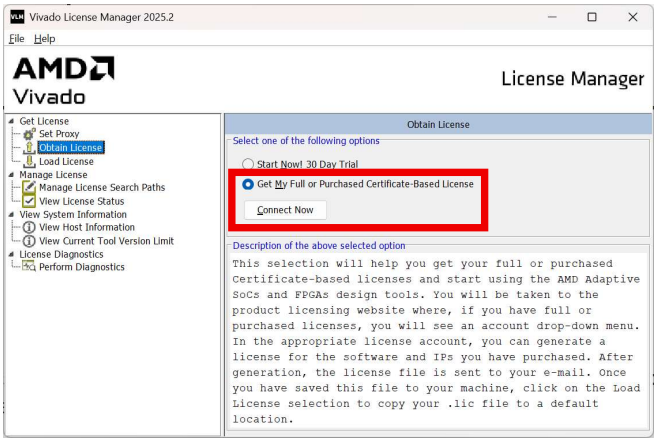


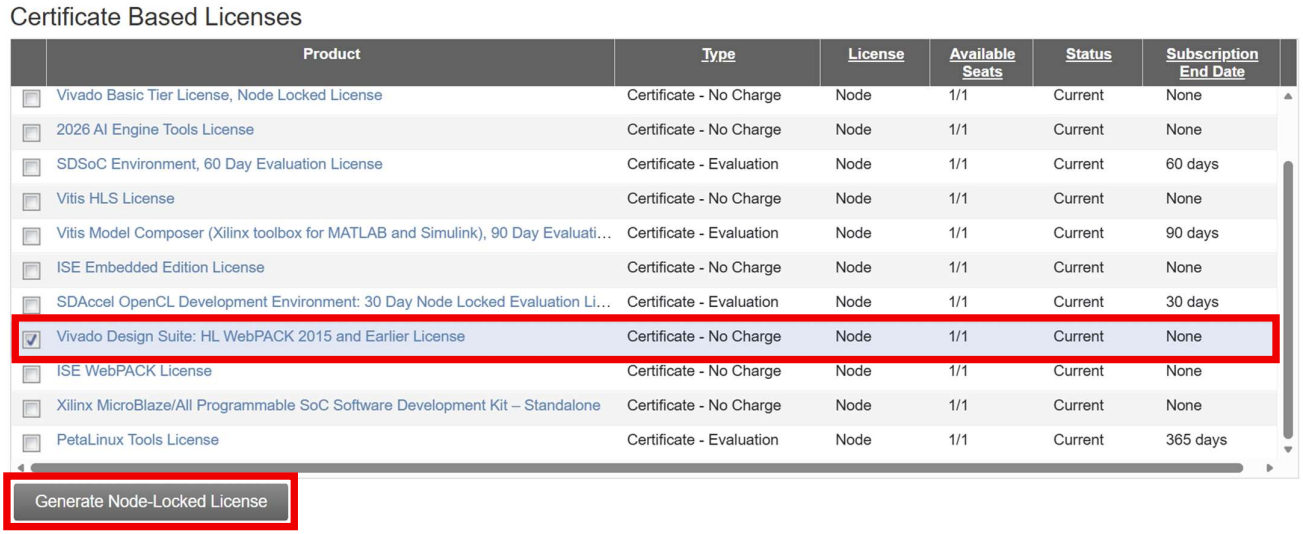




Software License

License the software by using the Vivado License Manager 2025.2:





DESCRIPTION OF THE SIMPLE BOOLEAN DESIGN TASK

The following task is required to be implemented on the Basys 3 development board:

- When switch **A** turns on, only **LED1** lights up.
- When switch **B** turns on, only **LED2** lights up.
- When both switches **A** and **B** turn on, **LED1**, **LED2**, and **LED3** light up.



UNDERSTANDING | TASK 1

Complete the truth table for the simple boolean design task:

INPUT		OUTPUT			MINTERM
A	B	LED1	LED2	LED3	
0	0				$\bar{A}\bar{B}$
0	1				$\bar{A}B$
1	0				$A\bar{B}$
1	1				AB

Deriving an SOP Boolean Equation for the Design Task

Given any truth table with any number of input variables, the sum-of-products (SOP) or product-of-sums (POS) form may be used to write out a Boolean equation for each output variable. Let us use the canonical SOP form for **LED1**:

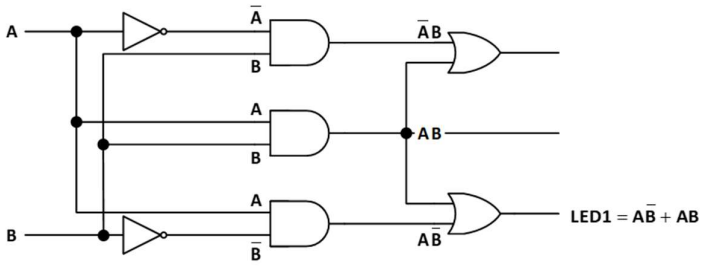
$$\mathbf{LED1} = A\bar{B} + AB$$

UNDERSTANDING | TASK 2

Work out the canonical SOP Boolean equations for **LED2** and **LED3**.

Illustrating Logic Expressions by Using a Schematic of Gates

The Boolean equations for **LED1**, **LED2**, and **LED3**, can be implement by using: 2 **NOT** gates, 3 **AND** gates, 2 **OR** gates



Verilog Hardware Description and FPGA Implementation

The Vivado software is an integrated design environment, among which we will be introducing the following:

- HDL Coding.
- Simulation of the design.
- Synthesis of the codes.
- Implementation of the design to a target FPGA.
- Programming an FPGA with a bitstream.

The remaining part of this lab manual will now briefly show the general steps required to go from the design task till programming the FPGA on the Basys 3 development board.

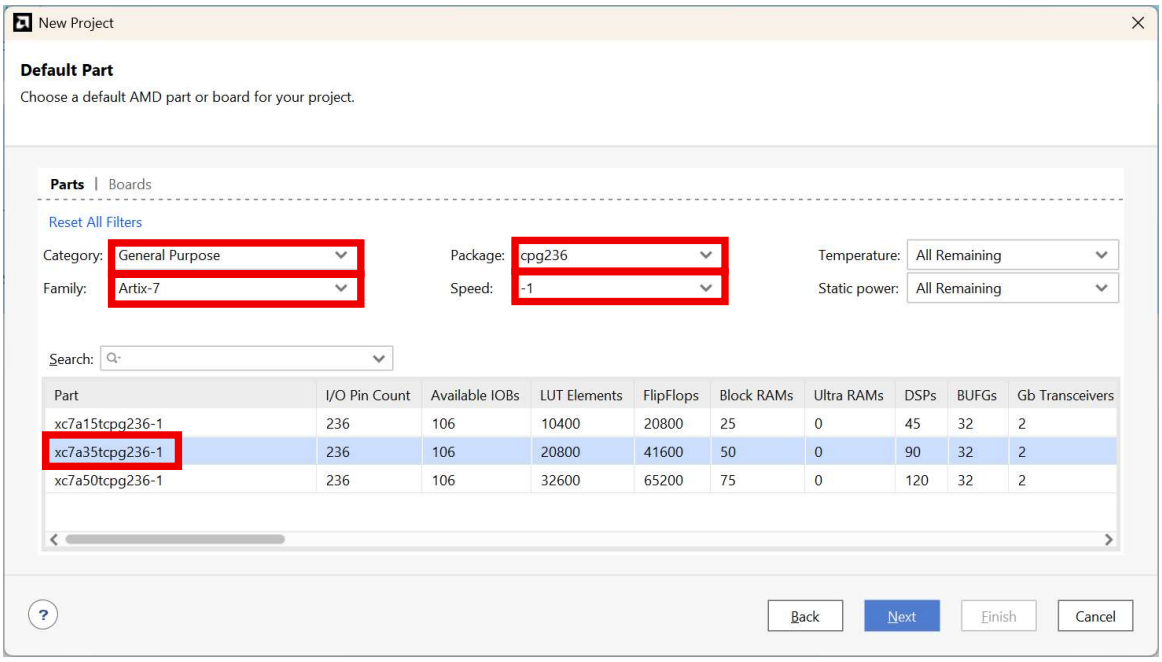
INTRODUCTORY QUICK START GUIDE TO XILINX'S VIVADO 2018.2 SOFTWARE

During your lab session, your EE2026 lab assistants may provide you helpful hints on the usage of the Vivado 2025.2 software, beyond the most basic things that are described in this section.

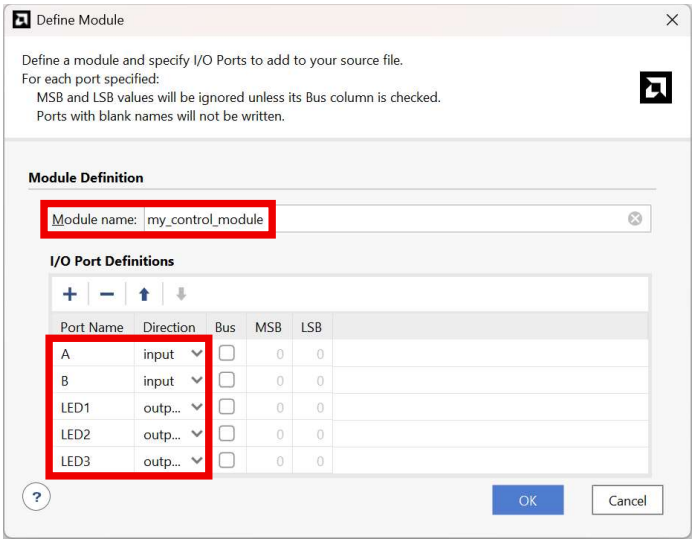
Creating a New Verilog Project in Vivado

- Open the **Vivado 2025.2** software.
- **Create Project**, then **Next**.
- Enter a **Project name** and **Project Location**.
  Ensure that the **Project name** and complete **Project location** for your project folder does not have any spaces, special characters, non-English characters, and that your **Project name** does not start with a number.
- Select **RTL Project** (Sub-options should be unchecked).
- **Create File.** File type is Verilog. File name example: simple_boolean.
- In **Add constraints**, click on **Next** without any changes.
- Ensure the correct chip is used. EE2026 uses the following FPGA chip:

  xc7a35tcpg236-1



- In the final summary window, click **Finish** to create the project.
- Create a module based on the inputs and outputs of the simple Boolean design task:



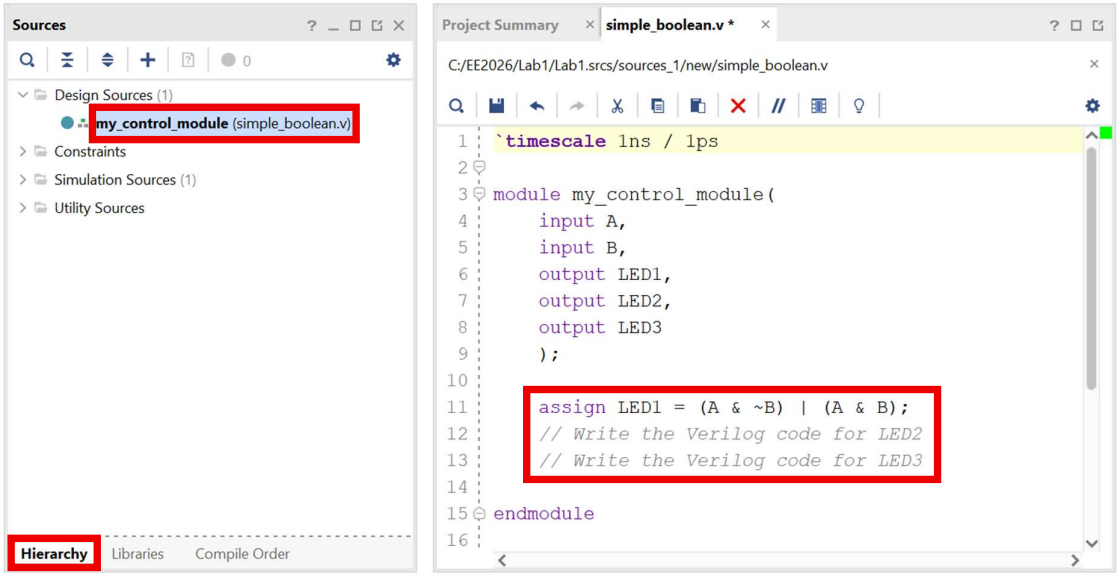
Writing Verilog HDL Code

Open the module that has been created by double clicking on it in the Sources window

UNDERSTANDING | TASK 3

Code the behaviour of the module by converting the SOP expressions for **LED1**, **LED2**, and **LED3** to the Verilog equivalent. The codes are to be inserted between "**module**" and "**endmodule**".

Common Operators		Verilog Representation
OR	$\mathbf{A + B}$	|
AND	$\mathbf{AB}$	&
NOT	$\mathbf{\bar{A}}$	~
XOR	$\mathbf{A \oplus B}$	^

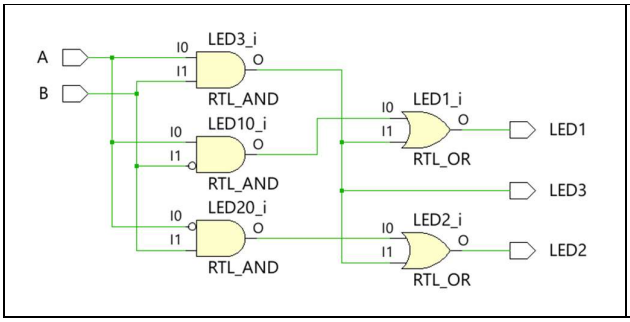


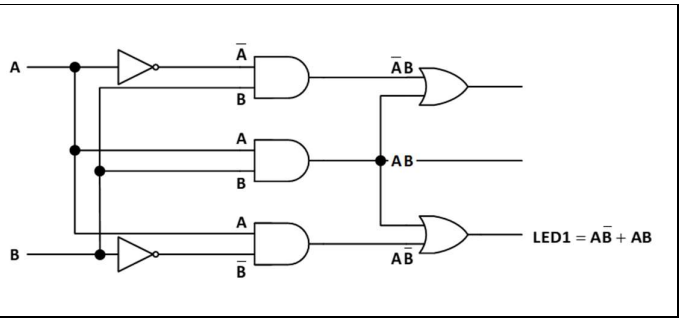
The **assign** statement causes the left-hand side of the expression to be updated *every* time there is a change on the right-hand side of the expression. It is called a *continuous assignment* statement, describing combinational logic whereby the output on the left is a function of current inputs on the right. All the **assign** statements run concurrently.

After writing the codes for LED1, LED2 and LED3, in the Flow Navigator (Left side window) expand **RTL ANALYSIS**: **Open Elaborated Design**, then select **Schematic. T**he Register Transfer Level (RTL) schematic of the design will appear.

UNDERSTANDING | TASK 4

What similarities and differences do you notice between the RTL schematic and the schematic from the previous section?



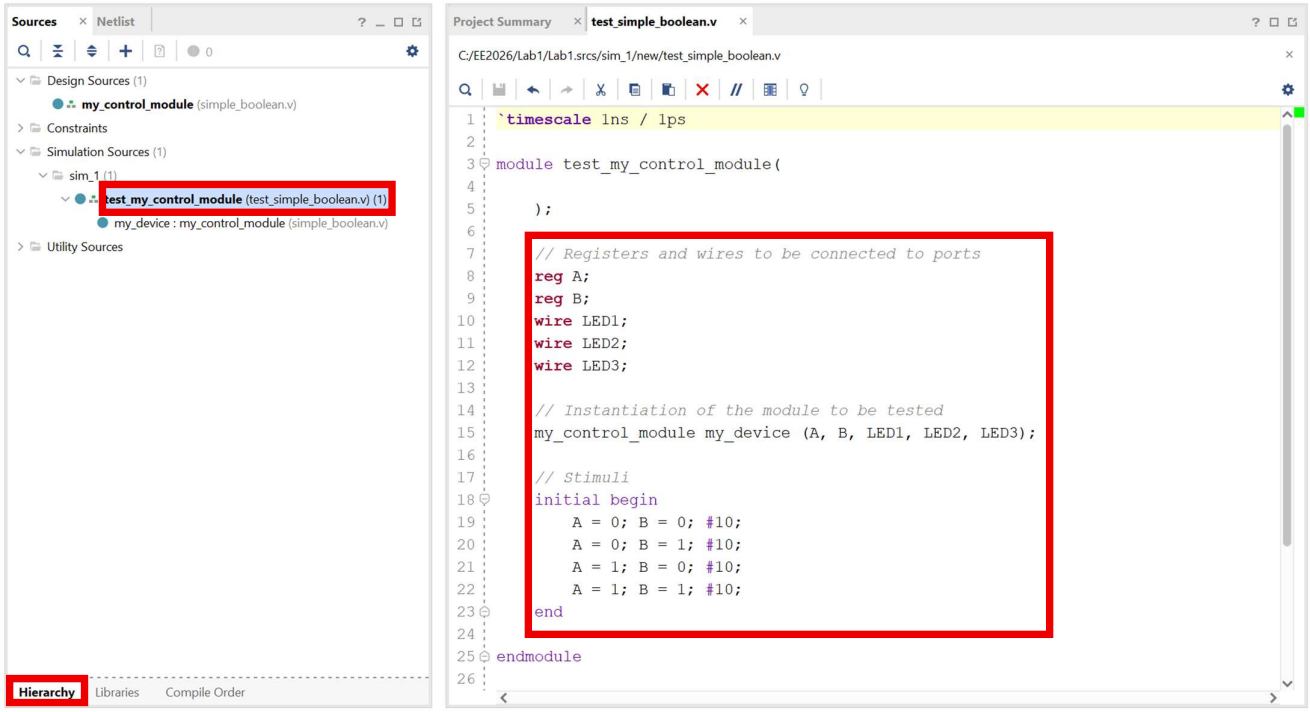


Testbench and Behavioural Simulation

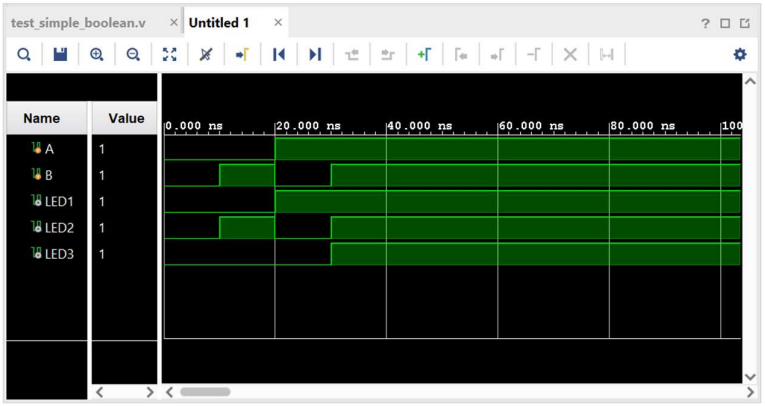
A testbench is an HDL module that is used to test another module:

- In the Flow Navigator, click on **PROJECT MANAGER -> Add Sources -> Add or create simulation sources**.
- **Create File** and provide a Verilog file name (Example: **test_simple_boolean**).
- In the subsequent **Define Module** window, provide a **Module name (**Example: **test_my_control_module**).
- Do not input any **I/O Port Definitions**, and click on **OK** to finish creating the simulation module template.

In the simulation file, write the code as shown below:



In the **Flow Navigator** window, click on **SIMULATION -> Run Simulation -> Run Behavioural Simulation**.
A noticeable waveform pattern may not be seen by default. From the menu, select **View → Zoom Fit**, or press **Ctrl+0**
How do the waveforms show that your design is indeed working as desired?

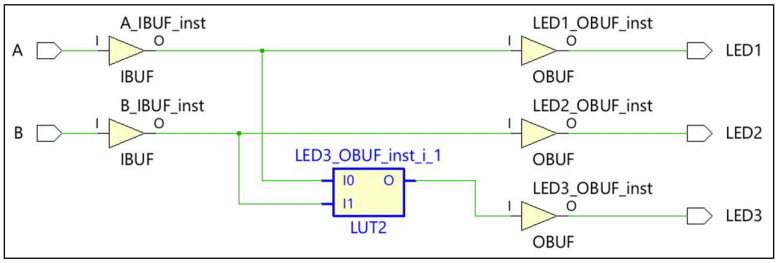


Suggestion: *Do not save the simulation window waveform, as this may consume a large amount of storage space.*

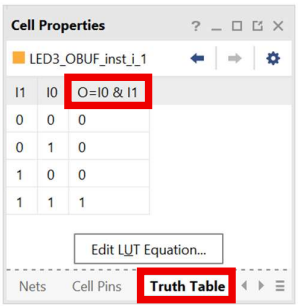
Synthesis

Logic synthesis transforms HDL code into an optimised set of logic gates to reduce the amount of hardware, and to efficiently perform the intended function.

- Right-click on your Verilog design source file and select **Set as Top**. (This is disabled if the file is already the top module) When there are multiple design and simulation modules, the "Set as Top" option selects the design, or simulation, modules to be considered when performing the different stages of the project flow.
- In the **Flow Navigator** window, click on **SYNTHESIS -> Run Synthesis** and wait for its completion.
  The Project Status Bar at the top right provides an indication of the ongoing progress.
- In the **Flow Navigator** window, click on **SYNTHESIS -> Open Synthesized Design -> Schematics** to see the following:

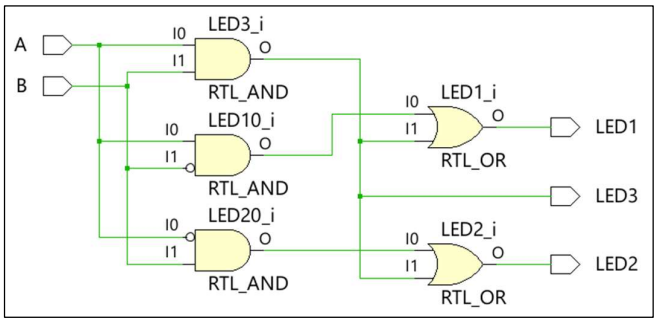


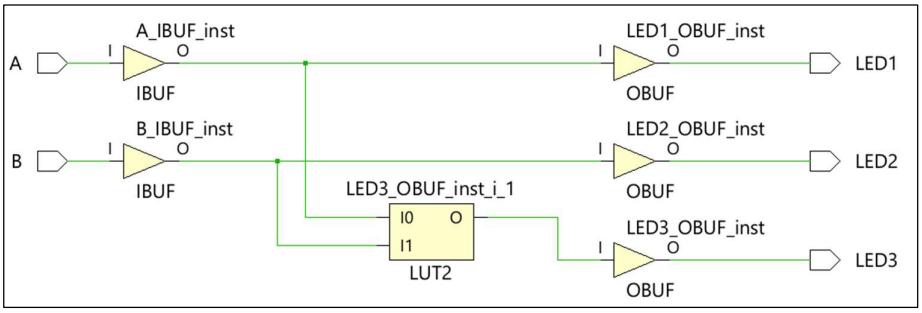
- Click on "LUT2" to select it. The Cell Properties allows the truth table for this LUT (AND gate in this example) to be viewed:



UNDERSTANDING | TASK 5

Compare the two schematics: Non-optimised (Schematics from RTL Analysis) and optimised (Schematics from Synthesis) and. How is this optimised circuit equivalent to the SOP equations of the simple boolean design task?

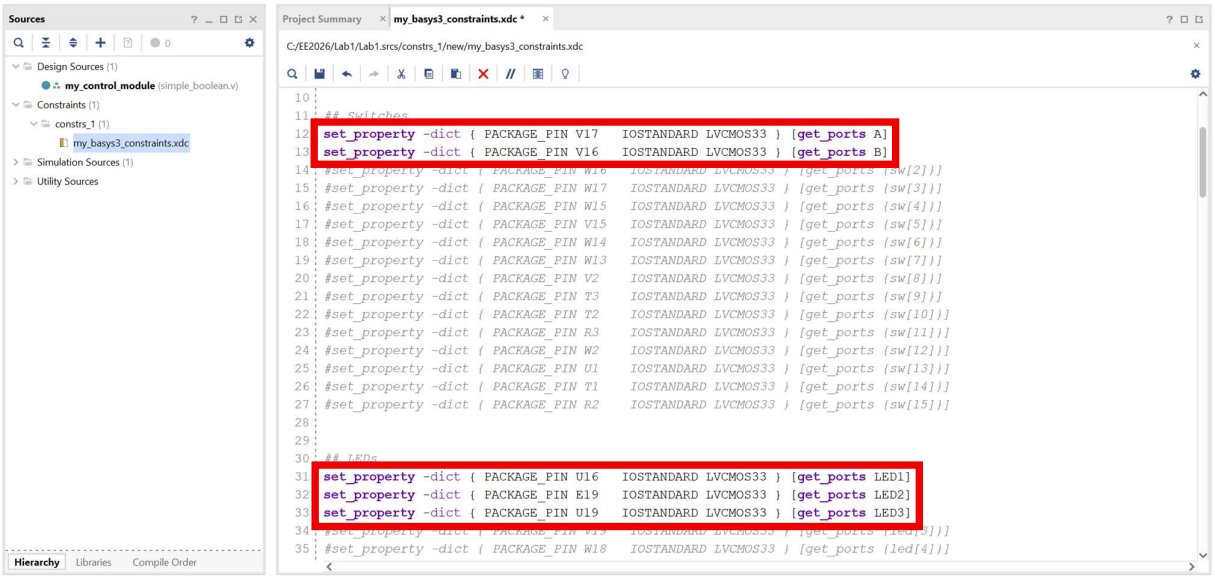




Design Constraints

Design constraints, such as physical I/O pin mapping, must be defined before doing an implementation:

- In the Flow Navigator, click on **PROJECT MANAGER -> Add Sources -> Add or create constraints**.
- **Create File** and provide an XDC (Xilinx Design Constraints) file name (Example: **my_basys3_constraints**).
- Open the **my_basys3_constraints.xdc** file from the Sources window.
- A template, known as the **Basys3_Master.xdc** is provided on CANVAS. Open that template using a basic text editor, such as notepad. Copy all the contents of the template to the my_basys3_constraints in Vivado.
- Link all signals (A, B, LED1, LED2, LED3) of your design to some physical pins of the FPGA, by making these changes:



Implementation, Bitstream Generation and Program Download

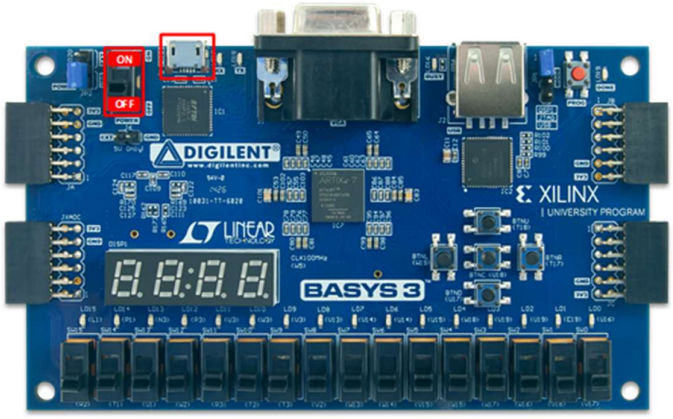
- In the Flow Navigator, click on **IMPLEMENTATION -> Run Implementation** and wait for its completion.
- In the Flow Navigator, click on **PROGRAM AND DEBUG -> Generate Bitstream** and wait for its completion.

Before using your Basys 3 development board, and to prevent potential damage to it, take note of the following recommendations and warnings to extend the longevity of the device:

⚠ Make sure the Basys 3 development board is powered OFF by placing SW16 (Top Left corner of the board) in the OFF position before connection to/removal from the USB port of the computer.

⚠ The chips on the board are electrostatic sensitive. Avoid touching them. Handle the board by the edges to prevent damage.

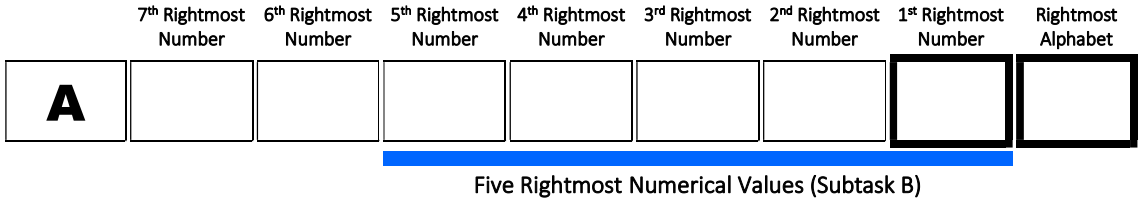
⚠ Make sure the board is not in contact with any metal components, whether above or below. Do not place any liquid sources near the FPGA board.



- Turn on the power to the Basys3 development board by setting SW16 to ON
- In the Flow Navigator, click on **PROGRAM AND DEBUG -> Open Hardware Manager -> Open target -> Auto Connect**
- After connection, click on **PROGRAM AND DEBUG -> Open Hardware Manager -> Program Device -> xc7a35t_0**
- By default, if the bitstream was successfully generated, the path name for the **Bitstream file** is automatically provided
- Click on **Program**. You may now test your design on the Basys3 by toggling SW0 and SW1

BEFORE STARTING ON GPT-1:

Each student has a personalised requirement based on student matriculation number. Carefully write down and use your exact number, as mistakes in the student matriculation number are not accepted. Appeals due to wrong number used for your personalised requirement will be rejected.



Students are required to carefully read all the assignment requirements, and to enhance their understanding by looking at the example.

Marks are given for using your correct student matriculation number, and grading is done only once. Re-submissions, late submissions, or updates, even if very simple or minor, are **not accepted** after the grading.

GRADED PRACTICE TASK 1 (GPT-1)

Complete as much as possible, in **ONE working bitstream for this whole assignment**. It is much better to have a working program with some completed functionalities, instead of submitting a program without a working bitstream (No marks given).

IMPORTANT CHARACTERS

In this assignment, these are the important characters to note from your student matriculation number:

- The rightmost alphabet of your student matriculation number (Initialistion)
- The five rightmost numerical values of your student matriculation number (Subtask B)
- The 1st rightmost numerical value of your student matriculation number (Subtask B)

INITIALISATION

When the program starts, all 16 active-high switches (SW0 to SW15) are in the OFF position. All 16 active-high LEDs (LD0 to LD15) are also OFF. The seven segment displays must automatically display the **rightmost alphabet** of your student matriculation number on all the 4 (four) anodes of the 7-segment displays. The character must be displayed **exactly as indicated** (Decimal point is OFF):

Rightmost Alphabet	A	B	E	H	J	L	M	N	R	U	W	X	Y
Required 7-Segments Character													

SUBTASK A

Consider the 10 (ten) switches SW0 to SW9 (Ignore SW10 to SW15). Whenever any of these 10 switches are ON, the corresponding LED LD**X**, where **X** is a number ranging from 0 to 9, must be ON. (For example, if SW0 is ON, then LD0 must be ON. If SW3, SW7 and SW9 are ON, then LD3, LD7 and LD9 must be ON.) **Do not put constraint to SW10 to SW15, and LD10 to LD15.**

SUBTASK B

Continuing from SUBTASK A, create your personal student matriculation number password based on the **five rightmost numerical values of your student matriculation number**.

These five digits (May be less than five digits if you have duplicate numbers) will represent the switches that need to be ON, while all the other switches between SW0 to SW9 must be OFF, to be considered a correct password. Whenever the password is wrong, all the 4 (four) anodes of the 7-segment displays show the **rightmost alphabet**. (For example, if all the ten switches SW0 to SW9 are turned ON, that is considered as a wrong password)

Whenever the password entered by the user is the correct password, then instead of all the 4 (four) anodes of the 7-segment displays showing your rightmost alphabet, only specific anodes of the 7-segment displays must show that alphabet. The specific anodes on which the character should be displayed is dependent on the **1st rightmost numerical value** of your student matriculation number, as indicated in the table below:

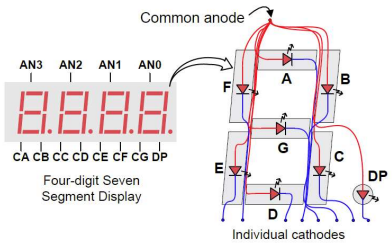
1st Rightmost Numerical Value	Anode AN3	Anode AN2	Anode AN1	Anode AN0
0 , 1	**ON**	OFF	OFF	**ON**
2 , 3	**ON**	OFF	**ON**	OFF
4 , 5	**ON**	**ON**	OFF	OFF
6 , 7	OFF	**ON**	**ON**	OFF
8 , 9	OFF	**ON**	OFF	**ON**

Note: *Marks will **not be given** if wrong switch constraints are used! If your student matriculation number password has a **0**, then it refers to SW**0**. If your student matriculation number password has a **1**, then it refers to SW**1**. If your student matriculation number password has a **2**, then it refers to SW**2** etc…Ensure you have noted down your 5 rightmost numerical values of your student matriculation number accurately for the password.*

SUGGESTIONS

- Create a new Vivado project for this assignment, instead of continuing from your previous Vivado project.
- This assignment can be **fully completed by using SOP / POS expressions**.
- This is a warm-up exercise. It is **not recommended to use contents not taught in this lab session**.
- Simulations are not required in the submission, but you are encouraged to practice them!

GETTING STARTED WITH THE SEVEN-SEGMENT DISPLAYS



Each segment behaves like an LED, and each controlled by the pin indicated in the **Basys3_Master.xdc**:

```
#7 Segment Display
set_property -dict { PACKAGE_PIN W7   IOSTANDARD LVCMOS33 } [get_ports seg_A]
set_property -dict { PACKAGE_PIN W6   IOSTANDARD LVCMOS33 } [get_ports seg_B]
set_property -dict { PACKAGE_PIN U8   IOSTANDARD LVCMOS33 } [get_ports seg_C]
set_property -dict { PACKAGE_PIN V8   IOSTANDARD LVCMOS33 } [get_ports seg_D]
set_property -dict { PACKAGE_PIN U5   IOSTANDARD LVCMOS33 } [get_ports seg_E]
set_property -dict { PACKAGE_PIN V5   IOSTANDARD LVCMOS33 } [get_ports seg_F]
set_property -dict { PACKAGE_PIN U7   IOSTANDARD LVCMOS33 } [get_ports seg_G]
set_property -dict { PACKAGE_PIN V7   IOSTANDARD LVCMOS33 } [get_ports seg_DP]
```

Each anode is like an output that needs to be turned ON, and each is controlled by the pin indicated in the **Basys3_Master.xdc**:

```
set_property -dict { PACKAGE_PIN U2   IOSTANDARD LVCMOS33 } [get_ports AN0]
set_property -dict { PACKAGE_PIN U4   IOSTANDARD LVCMOS33 } [get_ports AN1]
set_property -dict { PACKAGE_PIN V4   IOSTANDARD LVCMOS33 } [get_ports AN2]
set_property -dict { PACKAGE_PIN W4   IOSTANDARD LVCMOS33 } [get_ports AN3]
```

- In your constraint file, it is compulsory to put constraints to the 8 segments (7 segments + decimal point) of the seven-segment display, and to the 4 anodes of the seven-segment display.
- **Segments or anodes that need to be OFF must explicitly be assigned a value of '1' in your Verilog codes. Do not leave the values "floating"!** For example, if the 'dot' segment is required to be off, assign a value of '1' to it.

The circuit for the seven-segment displays on the Basys3 board is such that:

- A value of '1' turns OFF the segment / anode
- A value of '0' turns ON the segment / anode

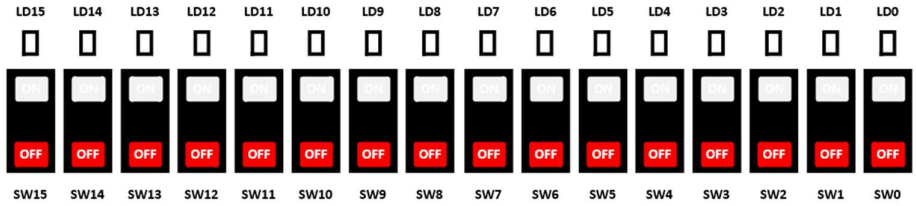
What devices / gates would you use to turn a '1' into '0', or vice-versa?

EXAMPLE OF GPT-1 BEHAVIOUR:

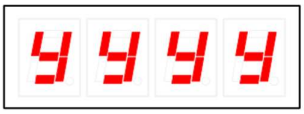
If your student matriculation number is A0159089Y, then:

1st rightmost numerical value:	**9** (Means only AN2 and AN0 will be ON if password is correct)
Five rightmost numerical values:	**59089** (Means password is SW0, SW5, SW8, SW9)
Rightmost alphabet:	**Y**

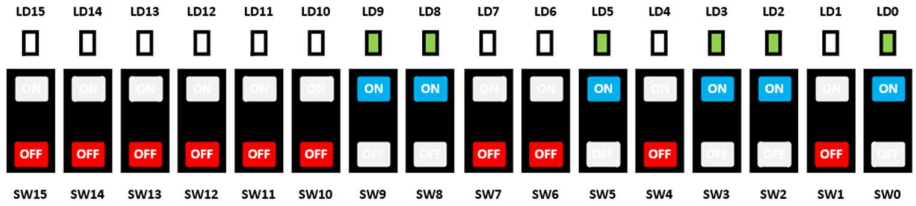
INITIALISATION



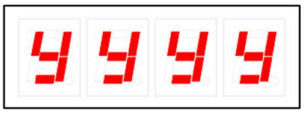
DISP1



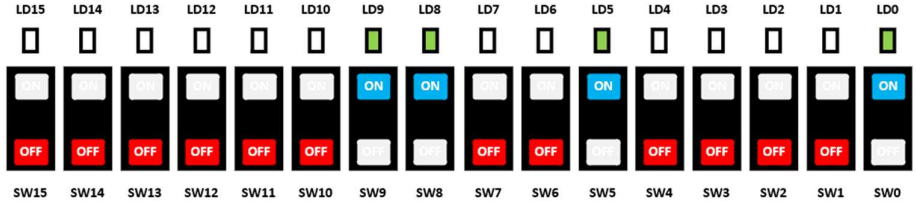
WRONG PASSWORD



DISP1



CORRECT PASSWORD



DISP1



CANVAS SUBMISSION INSTRUCTIONS

- Ensure that your bitstream has been successfully generated and tested on your Basys 3 development board **BEFORE** archiving your Vivado workspace for CANVAS upload.

- It is compulsory to archive your project in a compressed form without any saved simulation waveforms. Follow the instructions given in the pdf: "Archive Project in Vivado 2025.2". **The archive size should not exceed 5 MB in size for any lab assignments.** If the archive size exceeds, ensure that there is no ".sim" folder in your Vivado project folder before archiving.

- **After** following the instructions in "**Archive Project in Vivado 2025.2**", rename your project archive as indicated in the appendix of this lab manual.

- Upload to CANVAS EE2026 -> Assignments -> Lab 1 -> GPT-1 Submission (On-Time).

- Download your CANVAS archive after uploading. **Click and drag the single folder within that archive to desktop**, and then open the Vivado project (.xpr) in that **extracted folder** to see if it can be opened. You can ignore "out-of-date" warnings. **Check if you can also run your bitstream correctly.** No project file and no working bitstream is equivalent to losing all marks.

- GPT-1 is recommended to be completed within the lab session itself. You are however allowed to upload to CANVAS without penalties before **Monday 31st August 2026, 6:00 A.M.** Avoid planning to upload during the additional grace period of 2 hours. (Grace period is for last-minute cases, such as due to slow internet, wrong uploads etc.)

- The late submission folder closes 1 week after the original deadline. **Late submissions are not graded if submissions are found in the on-time folder. Do not submit in the late submission folder if you have already submitted in the on-time folder, as it will be ignored and only the on-time submission will be graded.** The late submission folder will be located at: CANVAS EE2026 -> Assignments -> Lab 1 -> GPT-1 Submission (Late).

- Re-submissions, late submissions, or updates, even if very simple or minor, are **not accepted** after your original submission has been graded.

- Submissions through emails, file sharing programs, or links, are **not accepted**. Submissions must be officially submitted to, and downloadable from, Canvas in **the correct on-time or late folder**.

Plagiarism is penalised with a 100% penalty for all SOURCES and RECIPIENTS

All past and future submissions, and marks, will be reviewed in greater detail, for any person found to have plagiarised

ALL THE SUBMISSION INSTRUCTIONS LISTED ABOVE WILL AFFECT YOUR GRADES!

GRADING PROCESS

Submissions not following all the ***CANVAS SUBMISSION INSTRUCTIONS*** (listed above) will not be graded immediately, and they will instead be reviewed towards the end of the semester. **You will not be able to see your submission results during the lab sessions in such situations**.

APPENDIX (COMPULSORY renaming just before CANVAS upload):

It is **compulsory to rename your project archive** just before the CANVAS upload, as listed in the table below.

Do not change any other part of the naming. Simply **copy** the naming from the table below, and **paste** it while renaming your project archive. Penalties will be incurred if your submission cannot be found according to the exact naming template below.

CANVAS may automatically add a suffix (Example. "-1", "-2", "-3" etc.) at the end of the filename if you submit multiple files. That is acceptable and we will grade the latest file submitted within that on-time folder.

Name	Archive Naming (.xpr.zip and .zip accepted)
AARON CHIN JIN HAO	GPT1_AARON CHIN JIN HAO_163_Archive
ADI RAMANLAL	GPT1_ADI RAMANLAL_033_Archive
ADVAIT BAGRI	GPT1_ADVAIT BAGRI_209_Archive
AHMED IMAAD	GPT1_AHMED IMAAD_686_Archive
ALYCIA YAK SHI YI	GPT1_ALYCIA YAK SHI YI_246_Archive
AMBAR	GPT1_AMBAR_974_Archive
ANDREW CHONG KIAN WEI	GPT1_ANDREW CHONG KIAN WEI_019_Archive
ANDREW YAP ZI ZEN	GPT1_ANDREW YAP ZI ZEN_117_Archive
ARNAV PARAG SALKADE	GPT1_ARNAV PARAG SALKADE_052_Archive
ARORA JITISHA	GPT1_ARORA JITISHA_189_Archive
ARUN PANCHAPAKESAN IYER	GPT1_ARUN PANCHAPAKESAN IYER_842_Archive
ATHARVA REGE	GPT1_ATHARVA REGE_841_Archive
BALAKRISHNAN SHRIKKANTH	GPT1_BALAKRISHNAN SHRIKKANTH_980_Archive
BARANIKUMAR SARITHA PADMANABAN	GPT1_BARANIKUMAR SARITHA PADMA_345_Archive
BETTS DANIEL AN-ZHE	GPT1_BETTS DANIEL ANZHE_421_Archive
BHAO TANUSH	GPT1_BHAO TANUSH_706_Archive
BI JIALIN	GPT1_BI JIALIN_083_Archive
BODAPATI VAMSHI KRISHNA	GPT1_BODAPATI VAMSHI KRISHNA_405_Archive
BRYAN CHUA XIANG YU	GPT1_BRYAN CHUA XIANG YU_710_Archive
CAI YI HANG	GPT1_CAI YI HANG_455_Archive
CHAN CHEE HENG	GPT1_CHAN CHEE HENG_721_Archive
CHANG YICHUN	GPT1_CHANG YICHUN_522_Archive
CHANG ZI SHYAN, SHERWIN	GPT1_CHANG ZI SHYAN SHERWIN_293_Archive
CHEN CHENG-SHIH	GPT1_CHEN CHENGSHIH_136_Archive
CHEN HOK LEONG	GPT1_CHEN HOK LEONG_601_Archive
CHEN ZHENONG	GPT1_CHEN ZHENONG_159_Archive
CHEW WAI JUN EVAN	GPT1_CHEW WAI JUN EVAN_524_Archive
CHIA MING HENG, TITUS	GPT1_CHIA MING HENG TITUS_200_Archive
CHUA CHOON HSIANG	GPT1_CHUA CHOON HSIANG_727_Archive
CHUA JINGHAO	GPT1_CHUA JINGHAO_532_Archive
DARREN TOH QING HONG	GPT1_DARREN TOH QING HONG_467_Archive
DONGJIN SHIN	GPT1_DONGJIN SHIN_719_Archive
DOUGLAS ETHAN KWOK	GPT1_DOUGLAS ETHAN KWOK_301_Archive
DUONG NGHIEP HOANG	GPT1_DUONG NGHIEP HOANG_675_Archive
EETHAN ZHANG YOU JIE	GPT1_EETHAN ZHANG YOU JIE_514_Archive
ELYSSA TJIN YI JING	GPT1_ELYSSA TJIN YI JING_142_Archive
ENZO HO	GPT1_ENZO HO_460_Archive
ERNEST GOH CHIN WEE	GPT1_ERNEST GOH CHIN WEE_869_Archive
EUGENE QUEK	GPT1_EUGENE QUEK_592_Archive
GABRIEL YU YAO SOON	GPT1_GABRIEL YU YAO SOON_978_Archive
GAN SIM RU DAYNA	GPT1_GAN SIM RU DAYNA_051_Archive
GANGADHAR NANDI GAUTHAM	GPT1_GANGADHAR NANDI GAUTHAM_850_Archive
GARG DIVYANSH	GPT1_GARG DIVYANSH_042_Archive

GARG SANCHIT AMIT	GPT1_GARG SANCHIT AMIT_679_Archive
GLAENNFJORD MILLA LINNEA ASTRID	GPT1_GLAENNFJORD MILLA LINNEA _gl_Archive
GLENNON PHUA QING XUN	GPT1_GLENNON PHUA QING XUN_609_Archive
GOH ROU HUI ASHLEY	GPT1_GOH ROU HUI ASHLEY_439_Archive
GOPINATHAN MAIYURI	GPT1_GOPINATHAN MAIYURI_060_Archive
GUPTA TANMAY	GPT1_GUPTA TANMAY_349_Archive
HAN QI-ZE	GPT1_HAN QIZE_320_Archive
HAN ZHONG DING	GPT1_HAN ZHONG DING_590_Archive
HENG YONG HEN	GPT1_HENG YONG HEN_977_Archive
HEU YU HUI	GPT1_HEU YU HUI_290_Archive
HUANG SHANGYI	GPT1_HUANG SHANGYI_564_Archive
IAN LEE JIAYI	GPT1_IAN LEE JIAYI_769_Archive
IFTIKHAR YUHAN BIN IRZAD YUHANIZAN	GPT1_IFTIKHAR YUHAN BIN IRZAD _693_Archive
ISAAC GOH YONG CHEE	GPT1_ISAAC GOH YONG CHEE_412_Archive
JAIN TANMAY	GPT1_JAIN TANMAY_216_Archive
JANG SEOK KEON	GPT1_JANG SEOK KEON_036_Archive
JAVIER LEE RUI YANG	GPT1_JAVIER LEE RUI YANG_587_Archive
JEREMIAH JAMES	GPT1_JEREMIAH JAMES_493_Archive
JORDAN RIATONO	GPT1_JORDAN RIATONO_175_Archive
JOSEFSSON CARL OSCAR WILLEHARD	GPT1_JOSEFSSON CARL OSCAR WILL_j_Archive
KANG KIAT YANG	GPT1_KANG KIAT YANG_549_Archive
KANNI KUMARAN SRISURAJ	GPT1_KANNI KUMARAN SRISURAJ_081_Archive
KARAN BONTHA	GPT1_KARAN BONTHA_031_Archive
KARTHIK SIVASUBRAMANIAN	GPT1_KARTHIK SIVASUBRAMANIAN_056_Archive
KARTIKEYAN RONITH	GPT1_KARTIKEYAN RONITH_987_Archive
KARUNANITHI NAVIN ANAND	GPT1_KARUNANITHI NAVIN ANAND_370_Archive
KAUSHAL RAJESH	GPT1_KAUSHAL RAJESH_862_Archive
KHOO KAI JUN	GPT1_KHOO KAI JUN_957_Archive
KIM DOYEON	GPT1_KIM DOYEON_415_Archive
KIM TAE-EUN	GPT1_KIM TAEEUN_919_Archive
KOH KA HOW	GPT1_KOH KA HOW_373_Archive
KOH LI WEI, OLIVER	GPT1_KOH LI WEI OLIVER_592_Archive
KOTHARI AADI	GPT1_KOTHARI AADI_808_Archive
KRONHAMN OSCAR LUDVIG	GPT1_KRONHAMN OSCAR LUDVIG_kr_Archive
KUMARESAN KUGAPRYAN	GPT1_KUMARESAN KUGAPRYAN_993_Archive
KYLE LEE YAO YU	GPT1_KYLE LEE YAO YU_663_Archive
LAM LOK SANG MATTHEW LUCAS	GPT1_LAM LOK SANG MATTHEW LUCA_294_Archive
LAM YAP TING	GPT1_LAM YAP TING_341_Archive
LAU ZHE WEN	GPT1_LAU ZHE WEN_969_Archive
LAW SHAO CHONG	GPT1_LAW SHAO CHONG_761_Archive
LEE CHANG FENG	GPT1_LEE CHANG FENG_463_Archive
LEE CHEONG WENG JUSTIN	GPT1_LEE CHEONG WENG JUSTIN_211_Archive
LEE EE XUAN	GPT1_LEE EE XUAN_562_Archive
LEE KWAN TZE	GPT1_LEE KWAN TZE_762_Archive
LEE PEI LING	GPT1_LEE PEI LING_692_Archive
LEE ZHON CHEN	GPT1_LEE ZHON CHEN_758_Archive
LEON MATTHEW WEI	GPT1_LEON MATTHEW WEI_323_Archive
LEROY KOH	GPT1_LEROY KOH_782_Archive
LEW YIK CHIN	GPT1_LEW YIK CHIN_716_Archive
LI MEIYI	GPT1_LI MEIYI_500_Archive
LI YI AN	GPT1_LI YI AN_037_Archive
LIAO YIRAN	GPT1_LIAO YIRAN_844_Archive
LIM CI ANN, ALOYSIUS	GPT1_LIM CI ANN ALOYSIUS_771_Archive

LIM JI YONG	GPT1_LIM JI YONG_851_Archive
LIM JING RUN	GPT1_LIM JING RUN_884_Archive
LIM JUN ZHAN	GPT1_LIM JUN ZHAN_890_Archive
LIM SHAU SHEE	GPT1_LIM SHAU SHEE_975_Archive
LIM SI HAN, GAVIN	GPT1_LIM SI HAN GAVIN_446_Archive
LIM ZI JING	GPT1_LIM ZI JING_395_Archive
LIU BOREN LAWRENCE	GPT1_LIU BOREN LAWRENCE_737_Archive
LOK WEI XIONG	GPT1_LOK WEI XIONG_376_Archive
LOKE JING JIE EVAN	GPT1_LOKE JING JIE EVAN_701_Archive
LOKHANDWALA ALIASGAR	GPT1_LOKHANDWALA ALIASGAR_001_Archive
LOO ZHONG EN SAMUEL	GPT1_LOO ZHONG EN SAMUEL_434_Archive
LOUISA CHUNG YU	GPT1_LOUISA CHUNG YU_654_Archive
LOW YI JIE SHERYL	GPT1_LOW YI JIE SHERYL_847_Archive
LYU JINHAN	GPT1_LYU JINHAN_276_Archive
MADHU KUMAR ANANNYA	GPT1_MADHU KUMAR ANANNYA_082_Archive
MADNANI KHUSHI	GPT1_MADNANI KHUSHI_069_Archive
MANDE NEIL ASHVINIKUMAR	GPT1_MANDE NEIL ASHVINIKUMAR_848_Archive
MANOJ AMIT	GPT1_MANOJ AMIT_044_Archive
MARCUS LEE WEI	GPT1_MARCUS LEE WEI_368_Archive
MARCUS LIM	GPT1_MARCUS LIM_380_Archive
MIRA KIM FABE	GPT1_MIRA KIM FABE_490_Archive
MOON JOONCHUL	GPT1_MOON JOONCHUL_280_Archive
MOON SI HO	GPT1_MOON SI HO_044_Archive
MUHAMMAD ASYRAF BIN ABDUL RAHIM	GPT1_MUHAMMAD ASYRAF BIN ABDUL_606_Archive
MUHAMMAD HAZIQ BIN ABDUL RAHIM	GPT1_MUHAMMAD HAZIQ BIN ABDUL _198_Archive
MUN JIA HAO	GPT1_MUN JIA HAO_276_Archive
MUTHUMANIKANTAN PAVITHRA	GPT1_MUTHUMANIKANTAN PAVITHRA_472_Archive
NANDHINI NACHIAPPAN	GPT1_NANDHINI NACHIAPPAN_089_Archive
NATARAJAN SRINIVASAN	GPT1_NATARAJAN SRINIVASAN_636_Archive
NATHAN BONG NENG QUAN	GPT1_NATHAN BONG NENG QUAN_650_Archive
NEHRU KABILLAN	GPT1_NEHRU KABILLAN_803_Archive
NEO KUANG YI JAYDEN	GPT1_NEO KUANG YI JAYDEN_533_Archive
NG YEE SUNG	GPT1_NG YEE SUNG_664_Archive
NG ZHI HOU	GPT1_NG ZHI HOU_719_Archive
NG ZI LE	GPT1_NG ZI LE_452_Archive
NGUYEN DUC KHOA	GPT1_NGUYEN DUC KHOA_243_Archive
NGUYEN TAN KIET	GPT1_NGUYEN TAN KIET_449_Archive
ONG JUNYIN, JUSTIN	GPT1_ONG JUNYIN JUSTIN_114_Archive
ONG LI KIAT, ISAAC	GPT1_ONG LI KIAT ISAAC_158_Archive
ONG MING YU	GPT1_ONG MING YU_773_Archive
OOI WEN YUE	GPT1_OOI WEN YUE_036_Archive
PAI SHASHANK LALITKUMAR	GPT1_PAI SHASHANK LALITKUMAR_208_Archive
PAN QI YUAN KEVIN	GPT1_PAN QI YUAN KEVIN_471_Archive
PATEL SIDDHARTH BUNTYBHAI	GPT1_PATEL SIDDHARTH BUNTYBHAI_689_Archive
PEMMASANI PRABAKARAN RASHMETA SAI	GPT1_PEMMASANI PRABAKARAN RASH_248_Archive
PENG BOHAO	GPT1_PENG BOHAO_755_Archive
PLOGNER OSKAR NILS ALFRED	GPT1_PLOGNER OSKAR NILS ALFRED__Archive
RAJASEKARAN KRITHIKA	GPT1_RAJASEKARAN KRITHIKA_539_Archive
RAMESH GOKUL SAI	GPT1_RAMESH GOKUL SAI_662_Archive
RAVI DEEPAK	GPT1_RAVI DEEPAK_148_Archive
REYAS MOHIDEEN FAWAZ	GPT1_REYAS MOHIDEEN FAWAZ_413_Archive
REYES JUSTIN NOEL DOTIMAS	GPT1_REYES JUSTIN NOEL DOTIMAS_680_Archive
ROYSTON CHEW YI DE	GPT1_ROYSTON CHEW YI DE_430_Archive

RUBENNISHWARAAN KARUPPIAH	GPT1_RUBENNISHWARAAN KARUPPIAH_691_Archive
RYAN PRIYANK SUNJAI DHAS	GPT1_RYAN PRIYANK SUNJAI DHAS_098_Archive
RYAN VERNADOS	GPT1_RYAN VERNADOS_024_Archive
SAM LEE HUEY SAMANTHA	GPT1_SAM LEE HUEY SAMANTHA_617_Archive
SAMPATHAWADUGE NILANA IMANDIV SILVA	GPT1_SAMPATHAWADUGE NILANA IMA_372_Archive
SELVAM ANAND	GPT1_SELVAM ANAND_304_Archive
SELVASINGH OLIVER JOSEPH	GPT1_SELVASINGH OLIVER JOSEPH_070_Archive
SEOW CHENG SI	GPT1_SEOW CHENG SI_802_Archive
SHEN YUNCONG	GPT1_SHEN YUNCONG_412_Archive
SHUKLA ATHARVA	GPT1_SHUKLA ATHARVA_251_Archive
SINGARAM KARTHIKEYAN	GPT1_SINGARAM KARTHIKEYAN_091_Archive
SPENCER TING TIEW YU	GPT1_SPENCER TING TIEW YU_254_Archive
SRIRAM ANUPAMA	GPT1_SRIRAM ANUPAMA_775_Archive
STEFAN CHOO BIN HAO	GPT1_STEFAN CHOO BIN HAO_098_Archive
STEVEN MARCUS KHOO TENG EE	GPT1_STEVEN MARCUS KHOO TENG E_978_Archive
SUA ISAAC	GPT1_SUA ISAAC_138_Archive
TAN BOON LEONG	GPT1_TAN BOON LEONG_443_Archive
TAN JIN XIANG	GPT1_TAN JIN XIANG_014_Archive
TAN KAI SHENG	GPT1_TAN KAI SHENG_237_Archive
TAN SWEE KIAT	GPT1_TAN SWEE KIAT_110_Archive
TAN YONG XIANG	GPT1_TAN YONG XIANG_557_Archive
TAN YU JUN	GPT1_TAN YU JUN_046_Archive
TEE REN WEY	GPT1_TEE REN WEY_983_Archive
THAM YOU JIE	GPT1_THAM YOU JIE_411_Archive
THOW JIA LE CALEB	GPT1_THOW JIA LE CALEB_252_Archive
TIFFANY HENG	GPT1_TIFFANY HENG_973_Archive
TIU SHENG YANG	GPT1_TIU SHENG YANG_815_Archive
TOH JIAN BIN	GPT1_TOH JIAN BIN_754_Archive
TOH SIEW HEAN	GPT1_TOH SIEW HEAN_512_Archive
TRAN HOANG KHANG	GPT1_TRAN HOANG KHANG_579_Archive
TRICIA LIM ZI NING	GPT1_TRICIA LIM ZI NING_575_Archive
TRIPATHI ANOUSHKA	GPT1_TRIPATHI ANOUSHKA_062_Archive
TRISTAN LAU YUE SHEN	GPT1_TRISTAN LAU YUE SHEN_950_Archive
TRUONG PHI HUNG	GPT1_TRUONG PHI HUNG_402_Archive
VISHWAK NATTAMAI	GPT1_VISHWAK NATTAMAI_645_Archive
VISHWANATH ANNANYA	GPT1_VISHWANATH ANNANYA_037_Archive
VITO BENEDICT HALIM	GPT1_VITO BENEDICT HALIM_783_Archive
VIVEKAN KATHIRAVAN	GPT1_VIVEKAN KATHIRAVAN_236_Archive
VUPPU NAGARAJAN YUVEN RAAJ	GPT1_VUPPU NAGARAJAN YUVEN RAA_686_Archive
WAHLGREN HANNA CHRISTINA	GPT1_WAHLGREN HANNA CHRISTINA__ch_Archive
WANG QICHEN	GPT1_WANG QICHEN_536_Archive
WILL CHIN JUN YU	GPT1_WILL CHIN JUN YU_806_Archive
WONG WEI LUN	GPT1_WONG WEI LUN_308_Archive
XANDER LIM	GPT1_XANDER LIM_790_Archive
XU RUIJIE	GPT1_XU RUIJIE_009_Archive
YEW SU YI	GPT1_YEW SU YI_674_Archive
YONG CHUAN ONN	GPT1_YONG CHUAN ONN_697_Archive
YU YIFAN	GPT1_YU YIFAN_811_Archive
YUE WEI AN JOSHUA	GPT1_YUE WEI AN JOSHUA_221_Archive
ZEPH TANG ZHENKAI	GPT1_ZEPH TANG ZHENKAI_461_Archive
ZHAO ZEYU	GPT1_ZHAO ZEYU_093_Archive
ZOE TAY HUI EN	GPT1_ZOE TAY HUI EN_198_Archive
ZOU YUYANG	GPT1_ZOU YUYANG_145_Archive

If you have registered for EE2026 labs late, your name may not be present in this lab manual. It will be included in subsequent lab manuals. Please use this naming template meanwhile: GPT1_*Name as on Matriculation Card*_Archive

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FINAL CHECKLIST FOR GPT-1:

Grading is done only once. Re-grading due to not meeting the checklist below are not accepted.

- ☐ I have used the **exact and correct numbers and alphabet** from my student matriculation number for the assignment.
- ☐ I have named the submitted archive **according to the given template**.
- ☐ The submitted archive is **less than 5 MB** (**Delete ".sim" folder if more than 5 MB.** Penalty of up to 10% apply if ≥5 MB).
- ☐ I confirm that the uploaded **submission is the correct / latest version**, and not an older one.
- ☐ I have **downloaded my submission** before the deadline and confirmed that it was properly uploaded to canvas.
- ☐ I confirm that the **grader need not "Generate bitstream"**, as the bitstream is already there during "Program Device".
- ☐ I have carefully verified the bitstream in my **CANVAS submission**, and it works exactly as I expected.